ATP yield from one molecule of glucose

STUDENT SHEET

Use your finished card layout to complete the table for **one molecule of blood glucose** that is fully oxidised. Fill in the first three rows, then calculate the ATP made by oxidative phosphorylation from your NADH and $FADH_2$ totals.

Assumptions to use

Each NADH gives about **2.5 ATP** and each $FADH_2$ gives about **1.5 ATP** (approximate P/O ratios, Hinkle, 2005).

Stage	ATP	NADH	$FADH_2$	CO_2
Glycolysis Glucose to 2 pyruvate				
PDH 2 pyruvate to 2 acetyl-CoA				
Krebs cycle 2 turns, one per acetyl-CoA				
Oxidative phosphorylation Electron transport chain				
Total per glucose				

Working for oxidative phosphorylation

NADH _____ × 2.5 = _____ ATP

$FADH_2$ _____ × 1.5 = _____ ATP

Total _____ ATP from oxidative phosphorylation

Extension question

Why does glucose from muscle glycogen give one more ATP than blood glucose?

Hint: compare the first step of each route into glucose-6-phosphate on your card layout.

Hinkle, P.C. (2005) 'P/O ratios of mitochondrial oxidative phosphorylation', *Biochimica et Biophysica Acta (BBA) Bioenergetics*, 1706(1-2), pp. 1-11.